

Mahmoud Mohanad Mahmoud

Protection engineer

 71-15 st- sedibashr-Alexandria-Egypt

SUMMARY

Senior protection engineer in power systems, proven expertise in developing protective relay schemes, performing fault analysis and generating precise relay settings for HV/EHV substations. Strong background in factory and site acceptance testing (FAT/SAT), Testing Panels such as GIS , Control and protection, using software programs to configure protection relays such as Siemens , ABB, GE and Schnider. work smart and hard to achieve great goals for my company .specialized in relay coordination for HV/EHV substation, achieving protection selectivity

EXPERIENCE

Protection engineer

Jan - 2025

Egyptian Electricity Transmission Company.

Project 1: Borg el Arab substation (500/220/66/22 KV)

Project 2 : NDHPS12 substation (220/66/11 KV)

Project 3 : NDHPS7 substation (220/66/11 KV)

Develop protection schemes for various substation equipment, including feeders, Transformers and buses for GIS, Control and protection Panels.

Check scheme for MV, HV and EHV systems such as AC,DC circuits, Interlock between CB, DS and ES and indication for them.

Testing Binary inputs ,outputs and LEDs for signals between GIS , Control and protection Panels from BCU and protection relays.

Use specialized software to calculate and configure relay settings from DIGSI 5 program for Siemens relays such as Differential relays (7UT82,85, 86), Distance relays such as (7SA87, 7SL86) and BCU such as (7SJ85,82,86, 6MD85,82,86) and S1 Agile P40 Program for GE relays such as Agile P443, P643, P543 and PCM 600 program for ABB relay such as REF615 relay

Check updates for Binary inputs , outputs , LEDs ,interlock from relay program then load Configuration to the device.

Testing protection relays functions for circuits and transformers such as Distance, Differential, over current and Earth fault by Omicron (CMC 356), Freja (Freja 300 ,500, Sverker (780, 900), CT analyzer and CPC 100

Testing CT loop, VT loop, by secondary injunction for current and voltages values then check theses values on Multimeter, BCU and protection relay to check if theses values according to the CT and VT ratio of equipment or not

Check CT's transformers for phases and neutral point by CT analyzer test to check Ratio, Resistance and Vknee point.

After scheme check for bays,injection primary current to check the loop of current on metering core and protection cores on Multimeter, BCU and protection relays are correct and no unbalance or any fault on the loop.

Check BI, BO, Trip contacts on scheme and protection relay

Check Topology by testing status of Busbar on DECSI 5 program by inject values of current on all bays for all sections and see them on the program, current of phases R, S,T, Differential and Restrain current.

Calculate values of pickup currents for all bays such as Transformers , Feeders, incomers of transformers. Testing Busbar protection relay (7SS85) by secondary current which is pickup value on all bays to make sure that all bays will be tripped only according to it's ratio current

Testing sensitivity for all bays by secondary injunction for max value of all bays which will be busbar rated or max current. Familiarity with communication protocols like IEC 61850, 618691-2

Integrated protection relays with SCADA system to enable real-time fault monitoring

Pre-commissioning engineer

Prepare dossiers of systems which include (Index, Remaining, MCC certificate, RFS certificate, check list of punches for construction, quality control, pre-commissioning, P&ID drawings and calibration certificates with QC team to clear punches of all Electrical & Instrument systems which on the first plan for signature MCC contact prepare for walkthrough by sending invitation by email to ENPPI for system which is finished construction and ready to be finish precommissioning . write draft punch list for Electrical or instrument device or equipment then signature it from PETROJET , ENPPI and GASCO who accepted all categories of punches (A,B,C), scope (PTJ, ENPPI, GASCO or Vendor) and Discipline (Electrical, Instrument, Civil or piping). Write list of punches then send it to ENPPI to review, signature and send to GASCO. Company. Send dossiers to ENPPI to pre-review it. write comments then send it to GASCO. After QC team clear all punches of category " A", complete dossier and send it to ENPPI to signature MCC certificate

Mar 2022 - Sep 2024**PETROJET. MIDOR (Middle East Oil Refinery)****Construction engineer**

Installing of explosion proof lighting system, convenience sockets, and power outlets as per standards.

Installing of power panels and cables for LV loads and HVAC system as per standards.

Supervision of contractor for cables, Trays, glands and connection. Inspection of equipments such as Lighting and HVAC panels, junction boxes ,heaters by secondary current injection, contact, isolation resistance, earth tester and continuity test. Testing and commissioning for cables, loads, CB, CT, and VT. Testing for circuit breakers of low and medium loads, VSD by megger Supervision of starting-up and energizing for LV and MV loads.

Measure a reading of vibration motor by vibration meter for low and medium motor (mm/s)

Good Knowledge of wiring, setting, data sheets, schematic diagram, single line diagram & layout of area.

Ensure that all engineering drawings and technical documents are per standards. Coordinate with other engineering teams to ensure all system

BMS engineer

Jan 2021 - Jan 2022

Advansys for trading and contracting. KAYAN 110 (OCTAGON)

Responsible for the engineering programming, graphic screens, building management, and control system in accordance with the project specifications and time frame, engineering adopted codes, and good engineering practices.

Composing your own software strategies and graphics engineering.

Perform physical wiring checks and plant start-ups, provide technical support to other trades and sub-contractors and carry out troubleshooting to diagnose system defects.

Excellent technical writing and communication skills. of buildings & Tunnels which have controllers which have (FACs, and I/o modules)by Display to take read of all items (Auto manual on/Off), STS (On/Off), S/S (start/stop), Alarm STS (Fault), Trip (Low speed or high-speed Trip).

Write reports of problems for any equipment then solve it.

Monitored building assets and operations through automated control systems, responding quickly to alarms and resolving technical issues within one hour on average for each incident.

Diagnosed complex problems related to HVAC controls, lighting automation & fire detection systems promptly. Thoroughly tested all new or upgraded software components prior to installation; successfully completed installations without any operational errors or down time while meeting customer deadlines every time.

EDUCATION

Electrical power engineering

Sep 2013 - Aug 2018

International academy for engineering and Media science

Project : security system bank by Arduino, how to control the bank by sensors such as level water sensor, motion sensor and heat sensor.

CERTIFICATES

Certificates from MIDOR Project

Certificate of experience from ENPPI Company for construction and commissioning phases of the project which are done successfully.

Certificate of appreciation from TECHNIP Energies Company for construction and commissioning work which is achieved by all teamwork.

SKILLS

Auto cad

Office

Software

configuration

Work hard

Communication skills

Team work

Software programs and scada system

Working under pressure

Troubleshooting

LANGUAGES

Arabic : Fluent

English : very good

Military status

Completed service : officer (3 years)